

Statistical and Computational Models

Lecture 1: Statistical and computational foundations

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Roadmap

The object of estimation

Estimating f

Loss and expected prediction error

What is 'Learning'?

Suppose we observe **pairs** $(Y_i, \mathbf{X}_i)$, drawn from a joint distribution $p(\mathbf{y}, \mathbf{x})$.

Our goal is to figure out the conditional $p(\mathbf{y} | \mathbf{x})$ for **unseen** pairs.

That is, we **learn** an **unobserved function**, using only observable data.

For example: Y_i is whether i voted, $\mathbf{X}_i$ is what we know about them, and $p(\mathbf{y} | \mathbf{x})$ tells us how the two are connected. We never observe $p(\mathbf{y} | \mathbf{x})$, instead we hope to learn it from a finite sample of pairs.

Question: is linear regression a form of statistical learning? Yes! We 'learn' the $\hat{\beta}$ that minimises the sum of squared residuals.

Estimand

An **estimand** is the quantity you would report if nothing was unobserved.

In the voting example, our estimand might be $\mathbb{P}(\text{vote} \mid \mathbf{x})$: the chance a voter with covariates $\mathbf{x}$ turns out to vote.

But it could be anything: $\mathbb{E}[Y \mid \mathbf{X} = \mathbf{x}]$, a coefficient β_j , a latent position, a topic proportion.

Good quantitative social science must start with the **estimand** first.

With a coherent estimand in place, you can then turn to questions of **estimation**.

Esti-what?

Estimand: the unobserved parameter or function we want.

↪ Here f , the turnout rate for some X : $\mathbb{P}(\text{vote} \mid \mathbf{x})$.

Estimator: a function we can apply to observed data.

↪ Here $\hat{f}$ – least squares, a cubic fit, a local average.

Estimate: a specific number the estimator returns.

↪ Here $\hat{f}(20) = 43.6$, the line's guess about turnout for those aged 20.

Remember: the estimand is fixed and unknown. The estimate changes with every sample.

Some vocabulary

Statistics	Machine learning
estimation	learning, training
covariate, regressor, independent variable	feature
outcome, response, dependent variable	label, target
model selection	tuning, model selection
out-of-sample prediction error	generalization error, test error
latent variable	representation, embedding

The same machinery is present in both literatures, but with very different vocabularies attached (helpful).

It is worth being familiar with both; we will sometimes switch back and forth.

Some setup

Back to the running example: we want to learn $p(y \mid \mathbf{x})$.

Assume we can rewrite this as a function with both systematic and random components:

$$Y = f(\mathbf{X}) + \varepsilon$$

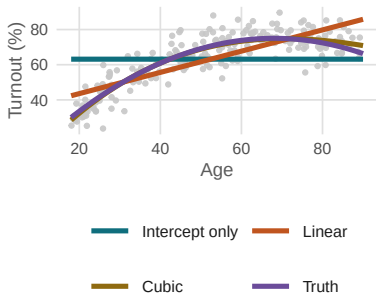
where f is fixed and unknown, and $\varepsilon \perp\!\!\!\perp \mathbf{X}$ and $\text{Var}(\varepsilon \mid \mathbf{X}) = \sigma^2$.

Read: Y is a **systematic function** of $\mathbf{X}$, plus **noise** that averages to zero.

Here $f(\text{age}) = \mathbb{P}(\text{vote} \mid \text{age})$: a binary Y means its conditional mean **is** the turnout rate.

Change $\mathbf{X}$ and you change Y via $f(\cdot)$. This may or may not be causal.

Models as abstractions of reality



An intercept-only model asserts turnout and age are unconnected.

A linear model asserts that turnout and age are monotonically and constantly connected.

A cubic model asserts a curved relationship, of a particular form.

Suppose we have 200 age-by-state units (e.g. one might be the turnout rate for 20-year olds in NY).

The truth can never be known (outside of a simulation), but it is what we are trying to learn.

Each fitted curve is a distinct **estimator** $\hat{f}$, returning its own **estimates** on a given draw.

Which model is 'best?'

That depends on how we evaluate them. Draw fresh observations from the same **DGP**, predict at each new $\mathbf{x}$, compare against the real $\mathbf{Y}$, and square. In expectation, at a single $\mathbf{x}$:

$$\begin{aligned}\mathbb{E}\left[(Y - \hat{f}(\mathbf{x}))^2 \mid \mathbf{X} = \mathbf{x}\right] &= \mathbb{E}\left[\left(f(\mathbf{x}) + \varepsilon - \hat{f}(\mathbf{x})\right)^2 \mid \mathbf{X} = \mathbf{x}\right] \\ &= \underbrace{\left[f(\mathbf{x}) - \hat{f}(\mathbf{x})\right]^2}_{\text{reducible}} + \underbrace{\sigma^2}_{\text{irreducible}}\end{aligned}$$

Cross term drops out: $\mathbb{E}[\varepsilon \mid \mathbf{X}] = 0$. Independence makes σ^2 the same at every $\mathbf{x}$.

Read: the first term is a property of the **estimator** you chose. The second is a property of the **data generating process**.

Which model is 'best?'

Let's do this with 2,000 fresh observations from our **DGP**.

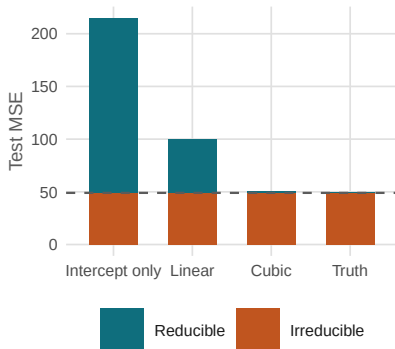
Predictor	Reducible	Irreducible	Test MSE
Intercept only	165.99	49	215.43
Linear	50.85	49	100.34
Cubic	1.34	49	49.73
Truth	0.00	49	48.58

The **irreducible** column does not change, it is σ^2 .

Note: the term **test MSE** is how we refer to the mean-squared error on a new, unseen draw from the DGP (the 'test set' – more soon).

Remember, we can only **split** the last column because this is a simulation – with real data you see **test MSE alone**.

The error floor



The dashed line is $\sigma^2 = 49$.

Going from the intercept to the cubic can shrink **reducible** error only.

The **irreducible** block is the same height everywhere.

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Two routes to $\hat{f}$

PARAMETRIC

Assume a form: $f(\mathbf{x}) = \beta_0 + \beta_1 \mathbf{x}$.

Estimate $p + 1$ parameters.

NON-PARAMETRIC

Assume only smoothness.

$\hat{f}(\mathbf{x})$ is a local average of nearby y .

We will spend some time over the next two weeks thinking about what you trade off when choosing between the two.

Supervised and unsupervised

SUPERVISED

Estimate $p(y | \mathbf{x})$ or $\mathbb{E}[Y | \mathbf{X}]$.

'Supervised' because the outcome constrains the answer. It **teaches**.

Continuous Y is **regression**.
Categorical Y is **classification**.

Evaluation is direct on the left: compare $\hat{y}$ to held-out y .

On the right it is indirect, and contested (uh oh).

Most of the course is **supervised** – in social science we usually observe Y – but we will take a few detours into the **unsupervised** world.

UNSUPERVISED

Learn features of $p(\mathbf{x})$ alone.

No outcome, so nothing constrains the answer.

Density estimation, dimension reduction, clustering, topics.

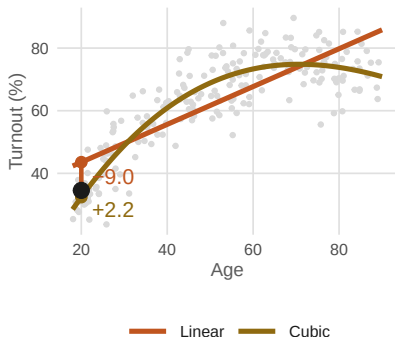
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Holding one observation out



Suppose we **hold out** one observation – the black dot. For this observation, turnout is 34.6%. We **did not fit** on it (that's what 'hold out' means).

The linear fit cannot bend down at the young end, so it **misses** high.

The cubic fit does bend, and misses by a fifth as much.

For two different models the same unit gives us two different **errors**.

Two predictions on one case

Predictor	Age	Turnout	Predicted	Raw	Squared	Absolute
Linear	20	34.6	43.6	-9.0	81.0	9.0
Cubic	20	34.6	32.4	2.2	4.6	2.2

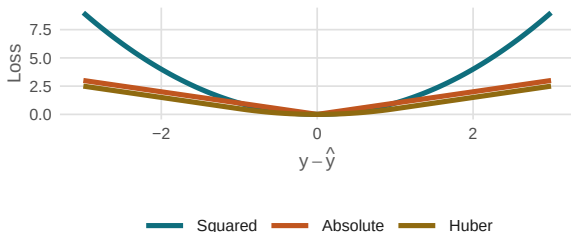
Let's try three different ways of counting the error: raw, squared, and absolute.

The raw error is not helpful – we don't know what the sign means (is a negative miss worse than a positive miss?).

The other two options rank the errors the same way (linear is worse than cubic), but the squared loss punishes a big miss harder than absolute loss does.

So the two costs **need not agree**.

Loss



$L(y, \hat{y})$ is the cost of reporting $\hat{y}$ when the truth is y .

Squared: $(y - \hat{y})^2$. Double the error, quadruple the cost.

Absolute: $|y - \hat{y}|$. Every unit of error costs the same.

Huber: squared near zero, absolute in the tails – a compromise.

Read: the loss function **prices the errors we necessarily make**. That choice is substantive, not a **computational convenience**.

Expected prediction error

$$\text{EPE}(f) = \mathbb{E}[L(Y, f(\mathbf{X}))] = \int L(y, f(\mathbf{x})) p(y, \mathbf{x}) dy d\mathbf{x}$$

$$f^* = \arg \min_f \text{EPE}(f)$$

Read: EPE is the average loss over the **data generating process**. f^* is the function that minimises it.

Note: ISL calls this the **expected test MSE** – squared loss, at one $\mathbf{x}$. Elsewhere you will meet it as the **risk**.

Expected prediction error

Under squared loss $f^*(\mathbf{x}) = \mathbb{E}[Y \mid \mathbf{X} = \mathbf{x}]$. Under absolute loss, it is the conditional median. Under 0–1 loss, it is the most probable class.

So the **estimand is a consequence of the loss**. You do not get to choose both.

While we **do not observe** the EPE, we can approximate it empirically, and minimise that instead.

Estimating EPE **in sample** causes problems, which we will address soon.

Training vs. test vs. EPE

Let's draw a very clear distinction early:

Quantity	Averages the loss over	Observable?
Training loss $\frac{1}{n} \sum_{i=1}^n L(y_i, \hat{f}(\mathbf{x}_i))$	the n observations we fit $\hat{f}$ on	always
Test loss $\frac{1}{m} \sum_{j=1}^m L(y_j, \hat{f}(\mathbf{x}_j))$	m observations not used in fitting $\hat{f}$	only if you have unseen data
EPE $\mathbb{E}[L(Y, \hat{f}(\mathbf{X}))]$	the population (or very many draws from the DGP)	never

Read: all three average the **same loss**, but differ in what they average it over.

We will come back to this very soon!

References

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